# URBAN HEAT ISLAND ANALYSIS

## Melbourne, Australia

Land Surface Temperature Mapping using Landsat 8 Thermal Remote Sensing

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| **Author** | Luke Kennedy |
| **Date** | June 2026 |
| **Scene Date** | 23 February 2020 (Late Summer) |
| **Satellite** | Landsat 8 OLI/TIRS — Band 4, 5, 10 |
| **Study Area** | Western & Central Greater Melbourne, Victoria, Australia |
| **Tools** | ArcGIS Pro (Spatial Analyst), USGS EarthExplorer |
| **LST Range** | 13.24°C – 37.80°C |

## Executive Summary

This report presents a Land Surface Temperature (LST) analysis of western and central Greater Melbourne derived from Landsat 8 Band 10 thermal infrared imagery acquired on 23 February 2020, representing Melbourne's late summer thermal peak. A six-step processing chain — from Top of Atmosphere Radiance through to emissivity-corrected LST — was implemented in ArcGIS Pro Spatial Analyst, producing spatially continuous surface temperature outputs at 30m resolution across the portion of Greater Melbourne captured within the Landsat 8 scene footprint (LC08_L1TP_092086_20200223). The eastern suburbs and outer fringe fall outside the scene extent and are therefore not included in this analysis.

Surface temperatures across the study area ranged from 13.24°C to 37.80°C. The analysis identifies a pronounced Urban Heat Island (UHI) effect concentrated in the city's western and northern growth corridors, where high impervious surface density and limited canopy cover drive surface temperatures significantly above regional averages. Port Phillip Bay and the northern forested areas — visible as the large cool zone in the upper portion of the scene — function as primary thermal buffers, registering the lowest LST values. NDVI analysis confirms a strong inverse relationship between vegetation cover and surface temperature, directly supporting the case for targeted urban greening in Melbourne's hottest suburbs.

These findings are directly relevant to the City of Melbourne Urban Forest Strategy and the Victorian Urban Cooling Strategy, both of which identify surface temperature reduction as a key climate adaptation priority.

## 1. Study Area & Context

Greater Melbourne is Australia's second-largest city, home to approximately 5.2 million people across a metropolitan area of roughly 9,990 km². The city's temperate oceanic climate (Köppen Cfb) produces warm to hot summers with high fire weather risk, and Melbourne regularly records extreme heat events during January and February. The February 2020 scene used in this analysis captures late-summer thermal conditions representative of Melbourne's annual peak heat loading.

The spatial coverage of this analysis is defined by the footprint of Landsat 8 scene LC08_L1TP_092086_20200223, which covers the western, central, and northern portions of the Greater Melbourne GCCSA boundary. The eastern suburbs and the Dandenong Ranges fringe lie outside this scene extent and are not represented in the outputs. Despite this partial coverage, the scene captures the majority of Melbourne's dense urban fabric, including the CBD, inner suburbs, and the western and northern growth corridors most relevant to the urban heat island discussion.

Melbourne faces a well-documented and politically prominent urban heat challenge. The urban heat island effect — driven by the replacement of vegetation with roads, rooftops, and paved surfaces — amplifies surface temperatures in the city's western and northern growth corridors. This spatial inequality in heat exposure is a central concern of both the City of Melbourne Urban Forest Strategy (target: 40% canopy cover by 2040) and the Victorian Government's Urban Cooling Strategy.

## 2. Methodology

All imagery was sourced from the USGS EarthExplorer platform (Landsat Collection 2 Level-1, LC08_L1TP_092086_20200223). Bands 4, 5, and 10 were downloaded and clipped to the Greater Melbourne GCCSA boundary (ABS, 2021). Processing was conducted in ArcGIS Pro Spatial Analyst Raster Calculator. The following six-step chain was applied:

| Step | Output | Formula / Expression |
|---|---|---|
| 1 | TOA Radiance | $L = 0.0003342 \times B10 + 0.1$ |
| 2 | Brightness Temperature | $BT = 1321.0789 / Ln(774.8853 / L + 1) - 273.15$ |
| 3 | NDVI | $NDVI = (B5 - B4) / (B5 + B4)$ |
| 4 | Proportion of Vegetation | $Pv = Square((NDVI - NDVImin) / (NDVImax - NDVImin))$ |
| 5 | Land Surface Emissivity | $LSE = 0.004 \times Pv + 0.986$ |
| 6 | Land Surface Temperature | $LST = BT / (1 + (0.00115 \times BT / 1.4388) \times Ln(LSE))$ |

Thermal constants applied: $K1 = 774.8853$ W/(m²·sr·µm), $K2 = 1321.0789$ K (Landsat 8 Band 10 TIRS1). Band 11 was excluded in accordance with USGS guidance on Band 11 calibration uncertainty. All outputs were produced at 30m spatial resolution.

# 3. Results & Analysis

## 3.1 Land Surface Temperature

LST across the study area ranged from 13.24°C to 37.80°C in the February 2020 late-summer scene, a thermal range of 24.56°C within a single metropolitan footprint. The spatial distribution reveals a clear pattern of elevated heat in the west and northwest: suburbs including Sunshine, Werribee, and Melton exhibit the highest surface temperatures, driven by high impervious surface fractions, sparse street tree canopy, and limited proximity to cooling water bodies.

The most prominent thermal buffer visible in the scene is Port Phillip Bay, whose open water surface registers LST values well below the surrounding urban fabric — a coastal cooling effect with direct implications for the resilience of bayside communities. A second major cool zone is visible in the northern portion of the scene, corresponding to the forested areas north of Melbourne's urban boundary (likely Kinglake and Toolangi State Forest), where dense vegetation cover and higher elevation produce the lowest LST values in the scene.

The Melbourne CBD and inner-northern suburbs show elevated LST consistent with dense urban morphology, though the thermal signature is spatially heterogeneous due to the interspersed presence of parklands including Royal Park, Fitzroy Gardens, and Albert Park Lake. It should be noted that the eastern suburbs — including Kew, Hawthorn, and the Yarra Valley fringe — fall outside the scene footprint and their thermal characteristics are not captured in this analysis.

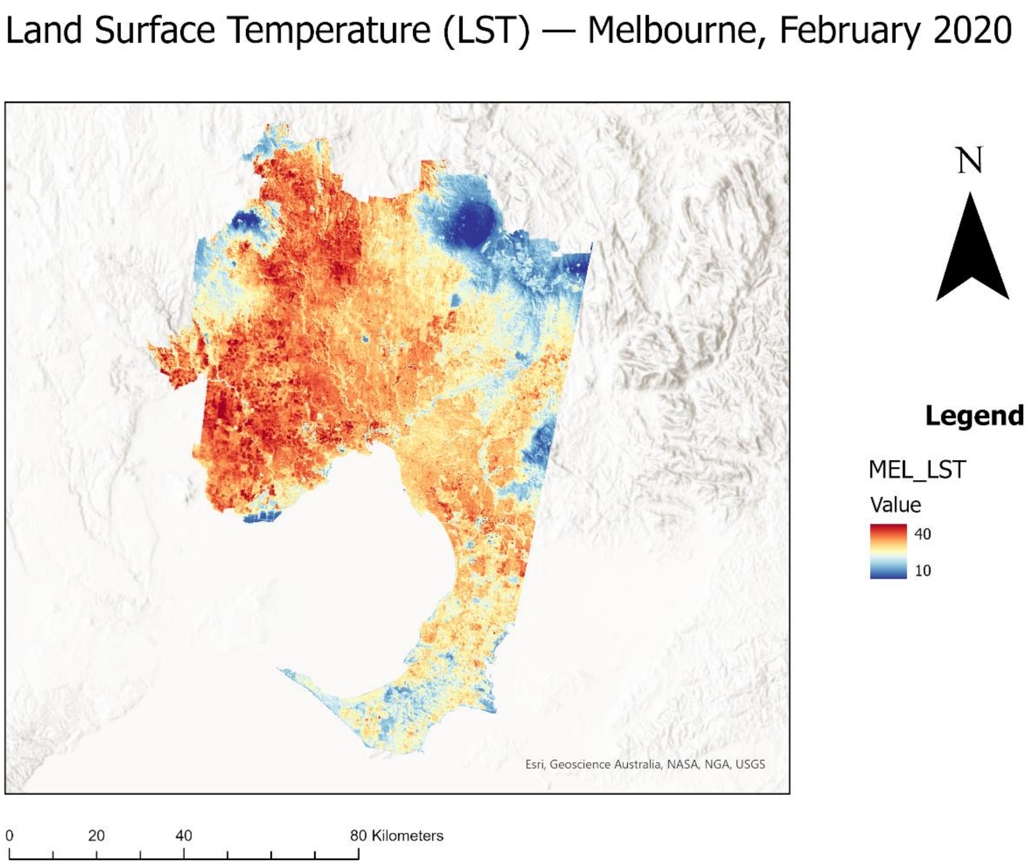


*Figure 1: Land Surface Temperature — Western & Central Greater Melbourne, February 2020 (13.24°C – 37.80°C)*

### 3.2 NDVI & Vegetation Cover

The NDVI layer confirms a strong spatial inverse relationship between vegetation fraction and surface temperature across the study area. Within the scene coverage, the highest NDVI values are recorded in the northern forested zone and along the Yarra River corridor through the inner suburbs — areas that correspond directly to the lowest LST values in the scene. The western growth corridor (Melton, Wyndham) displays characteristically low NDVI values, reflecting the dominance of new residential development with minimal established vegetation cover and high impervious surface exposure.

This NDVI-LST relationship propagates through the processing chain: higher NDVI yields higher Proportion of Vegetation (Pv), which elevates Land Surface Emissivity (LSE), which in turn reduces the emissivity-corrected LST output. The result is a measurable, spatially consistent cooling effect in vegetated zones that directly quantifies the thermal benefit of Melbourne's existing green infrastructure — and the cost of its absence in heat-exposed growth areas. Note that the Dandenong Ranges, frequently cited as Melbourne's eastern green lung, fall outside the scene footprint and their NDVI and LST characteristics are not captured here.

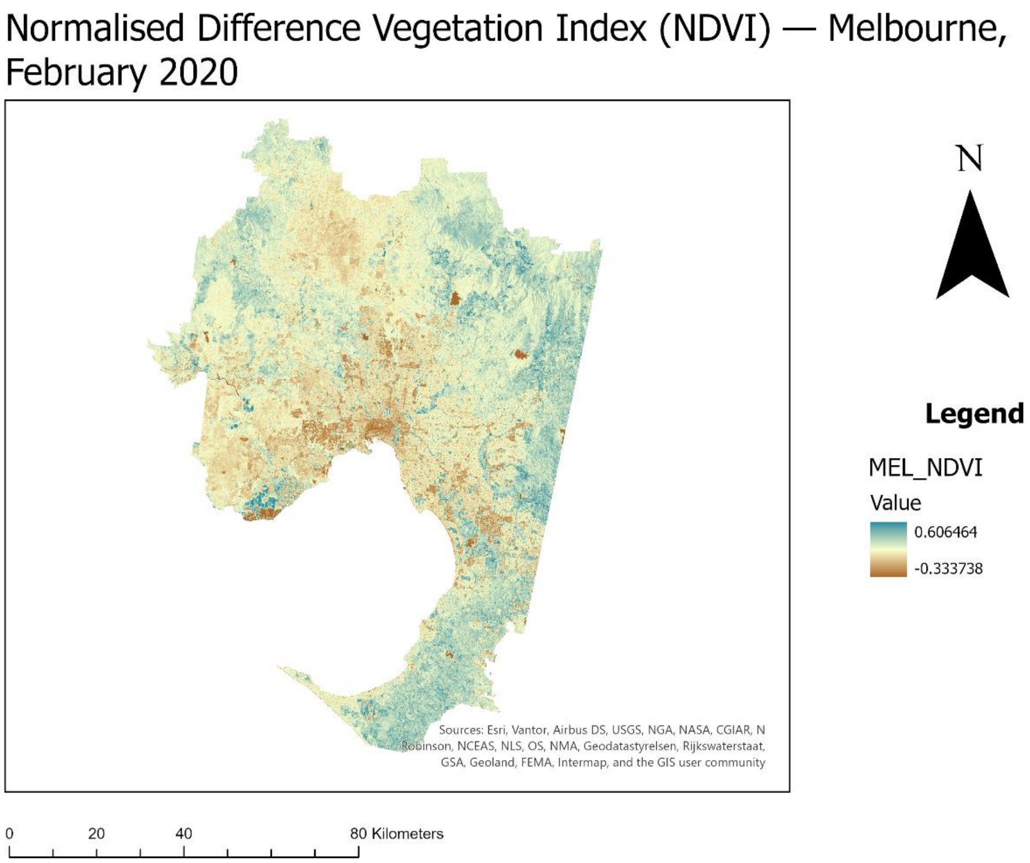


*Figure 2: Normalised Difference Vegetation Index (NDVI) — Western & Central Greater Melbourne, February 2020*

# 4. Key Findings

- LST ranged from 13.24°C to 37.80°C across the study area in February 2020, a thermal range of 24.56°C within the western and central portions of Greater Melbourne captured by the Landsat 8 scene.

- A pronounced west-northwest heat signature is evident: suburbs including Sunshine, Werribee, and Melton record the highest surface temperatures due to high impervious surface density and limited canopy cover.
- Port Phillip Bay exerts a clear coastal cooling effect, with LST values along the foreshore substantially below the surrounding urban fabric — the most spatially prominent thermal buffer in the scene.
- A large cool zone in the northern portion of the scene corresponds to forested areas north of Melbourne's urban boundary, where dense vegetation and elevation suppress surface temperatures to the lowest values in the dataset.
- NDVI is a strong inverse predictor of LST across the study area — suburbs and zones with higher vegetation fraction demonstrate measurably lower surface temperatures, quantifying the cooling benefit of urban green infrastructure.
- The Yarra River corridor produces a visible linear cooling effect through the inner suburbs, consistent with the latent heat buffering of open water and riparian vegetation.
- Eastern suburbs including Kew, Hawthorn, and the Dandenong Ranges fringe fall outside the scene footprint and are excluded from this analysis — a limitation to be addressed in future work using an adjacent scene tile.

## 5. Policy Implications

The spatial distribution of surface temperatures identified in this analysis has direct relevance to Melbourne's existing urban heat policy frameworks:

City of Melbourne Urban Forest Strategy (40% canopy cover target by 2040): This analysis provides spatial evidence of which areas within the scene coverage would benefit most from accelerated canopy investment. The western growth corridor — where surface temperatures are highest and NDVI is lowest — represents the priority zone for street tree planting and green infrastructure development.

Victorian Urban Cooling Strategy: The pronounced west-to-cool-north thermal pattern identified here supports the strategy's focus on heat equity — western Melbourne communities face disproportionately high surface heat exposure relative to areas with higher canopy cover and greater proximity to cooling water bodies and forested land.

Urban Growth Boundary Planning: The Melton and Wyndham growth areas show the highest LST values and lowest NDVI in the scene. As these areas continue to develop, mandating minimum canopy cover ratios and cool surface materials in planning approvals would directly address the thermal outcomes quantified in this analysis.

## 6. Limitations

- The Landsat 8 scene footprint (LC08_L1TP_092086_20200223) does not fully cover the Greater Melbourne GCCSA boundary. Eastern suburbs including Kew, Hawthorn, Frankston, and the Dandenong Ranges fringe fall outside the scene extent and are excluded from all outputs. A complete metropolitan analysis would require mosaicking an adjacent scene tile.
- Landsat 8 Band 10 has a native spatial resolution of 100m, resampled to 30m in the Level-1 product. This resampling introduces sub-pixel averaging that may smooth fine-scale thermal variation in heterogeneous urban environments.
- The single-date scene captures a snapshot of thermal conditions on one day in February 2020 and does not account for diurnal or inter-annual variability. A multi-temporal analysis would provide a more robust characterisation of Melbourne's heat exposure.
- The emissivity model applied ($LSE = 0.004 \times Pv + 0.986$) is a simplified two-endmember approach. More sophisticated emissivity retrieval methods (e.g. TES algorithm) may improve accuracy in spectrally complex urban environments.
- Atmospheric correction was not applied beyond the TOA Radiance conversion. Full atmospheric correction using MODTRAN or similar would improve absolute accuracy of derived LST values.

## Data Sources

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| **Landsat 8 Imagery** | USGS EarthExplorer — LC08_L1TP_092086_20200223 (Collection 2 Level-1) |
| **Study Area Boundary** | ABS Greater Capital City Statistical Areas (GCCSA) — Greater Melbourne, 2021 |

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| **Processing Software** | Esri ArcGIS Pro 3.x — Spatial Analyst Extension, Raster Calculator |
| **GitHub Repository** | github.com/lukek6892/urban-heat-island-melbourne-landsat8 |